Lesson 44: Average Reward MDPs

AI/ML Mathematics

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1 Motivation and Overview

Consider a data center scheduling agent that manages server workloads continuously, with no natural endpoint. The agent routes incoming requests to servers, balancing throughput against latency. Should the agent use a discount factor $\gamma = 0.99$? Or $\gamma = 0.999$? The choice of γ affects both the optimal policy and the convergence rate of algorithms, yet in many continuous-operation settings there is no intrinsic reason to discount future performance. The agent should maximize its *long-run average reward per step* – the rate of reward accumulation over an infinite horizon.

The average reward criterion eliminates the discount factor entirely, replacing the discounted return $\sum_{t=0}^{\infty} \gamma^t R(S_t, A_t)$ with the long-run average $\lim_{T\to\infty} \frac{1}{T} \sum_{t=0}^{T-1} R(S_t, A_t)$. This leads to a fundamentally different mathematical structure: the Bellman equation acquires a scalar offset (the *gain*), value functions are defined only up to an additive constant, and the analysis requires ergodicity conditions on the Markov chain. This lesson develops the theory of average reward MDPs, culminating in the gain-bias equations and a brief treatment of Blackwell optimality.

Why Average Reward MDPs Matter for RL/ML

- **Continuing tasks**: Many real-world RL applications (network routing, inventory management, robot operation) have no natural episode boundary; the average reward criterion is their natural objective.
- **R-learning**: The R-learning algorithm is the average-reward analogue of Q-learning, directly estimating the gain and relative values without a discount factor.
- **Sensitivity to γ**: Discounted algorithms suffer from sensitivity to the choice of γ; the average reward criterion avoids this hyperparameter entirely.
- **Blackwell optimality**: A Blackwell optimal policy is simultaneously optimal for all discount factors sufficiently close to 1, connecting discounted and average reward theory.

Prerequisites. MDP model, stationary policies, Bellman operators, contraction properties, value iteration, policy iteration (Lessons 39–43). Markov chains, stationary distributions, ergodicity (Lesson 12). Linear algebra: eigenvalues, matrix inverses, rank (Lessons 5–6).

Learning objectives. After completing this lesson, the student will be able to:

- *Define* the average reward criterion and the gain of a policy.
- *State* and *prove* the gain-bias (average reward Bellman) equations for unichain MDPs.
- *Classify* MDPs as unichain, multichain, or communicating, and *explain* why the classification matters.
- *Derive* the relationship between average reward and discounted reward as $\gamma \to 1$.
- *Define* Blackwell optimality and *state* Blackwell's theorem on the existence of Blackwell optimal policies.
- *Compute* the gain and bias for small unichain MDPs.

2 References

- Puterman [2014, Chapters 8–9] – Comprehensive treatment of average reward MDPs, gain-bias equations, unichain and multichain theory, and Blackwell optimality.

- Bertsekas [2012, Chapter 7] – Average cost dynamic programming, including the relative value iteration algorithm.
- Blackwell [1962] – Original paper proving the existence of discount-optimal policies that are simultaneously optimal for all large enough discount factors.
- Howard [1960] – Introduction of the gain and bias framework and policy iteration for average reward.
- Derman [1970] – Foundational treatment of average reward in finite Markov decision processes.

3 Markov Chain Classification

The average reward criterion requires careful analysis of the long-run behavior of Markov chains induced by policies. We begin by recalling the relevant classification.

Definition 3.1 (Induced Markov Chain)**.** A stationary policy π on an MDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma)$ induces a Markov chain with transition matrix $P^\pi \in \mathbb{R}^{S \times S}$, where $P^\pi_{ss'} = \sum_{a \in \mathcal{A}} \pi(a \mid s) P(s' \mid s, a)$.

Definition 3.2 (Communication Classes)**.** State s *communicates with* state s' (written $s \leftrightarrow s'$) if there exist integers $m, n \geq 1$ such that $(P^\pi)^m_{ss'} > 0$ and $(P^\pi)^n_{s's} > 0$. A *communication class* (or *recurrent class*) is a maximal set of mutually communicating states. States not in any recurrent class are *transient*.

Definition 3.3 (Chain Types)**.** A Markov chain with transition matrix P^π is:

1. *Irreducible* (or *ergodic*) if all states form a single communication class.
2. *Unichain* if it has exactly one recurrent class (plus possibly transient states).
3. *Multichain* if it has two or more recurrent classes.

An MDP is *unichain* if every deterministic stationary policy induces a unichain Markov chain. An MDP is *communicating* if for every pair of states s, s', there exists a deterministic stationary policy d such that s' is reachable from s under d.

Example 3.4 (Unichain vs. Multichain)**.** Consider $\mathcal{S} = \{1, 2, 3\}$ with two deterministic policies:

Policy d_1: $P^{d_1} = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0.5 & 0.5 \\ 1 & 0 & 0 \end{smallmatrix}\right)$. All states communicate (irreducible).

Policy d_2: $P^{d_2} = \left(\begin{smallmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{smallmatrix}\right)$. State 1 is an absorbing recurrent class; $\{2, 3\}$ is another recurrent class. This is multichain.

If these are the only two deterministic policies, the MDP is *not* unichain (since d_2 is multichain). However, it is communicating, because from state 1 we can reach states 2 and 3 under d_1.

Proposition 3.5 (Stationary Distribution for Unichain Policies)**.** *If policy π induces a unichain Markov chain, then there exists a unique stationary distribution $\mu^\pi \in \Delta(\mathcal{S})$ satisfying $(\mu^\pi)^\top P^\pi = (\mu^\pi)^\top$. Moreover, $\mu^\pi(s) > 0$ for all states s in the recurrent class, and $\mu^\pi(s) = 0$ for transient states.*

Proof. By the Perron-Frobenius theorem, a stochastic matrix with a single recurrent class has eigenvalue 1 with algebraic multiplicity 1. The corresponding left eigenvector, normalized to sum to 1, is the unique stationary distribution. States in the recurrent class have positive weight because the restriction of P^π to the recurrent class is irreducible. □

4 The Average Reward Criterion

Definition 4.1 (Average Reward / Gain)**.** The *gain* (or *average reward*) of a stationary policy π starting from state s is

$$\rho^\pi(s) = \liminf_{T\to\infty} \frac{1}{T}\mathbb{E}_\pi\left[\sum_{t=0}^{T-1} R(S_t, A_t) \;\middle|\; S_0 = s\right].$$

Remark 4.2. We use lim inf rather than lim because the limit may not exist for general (multi-chain) MDPs. For unichain MDPs, the limit exists and equals $\rho^\pi(s)$ for all s (see Theorem 4.3 below).

Theorem 4.3 (Gain of a Unichain Policy)**.** *If policy π induces a unichain Markov chain with stationary distribution μ^π, then the gain is independent of the initial state:*

$$\rho^\pi(s) = \rho^\pi := \sum_{s'\in\mathcal{S}} \mu^\pi(s') \sum_{a\in\mathcal{A}} \pi(a \mid s')R(s', a) = (\mu^\pi)^\top r^\pi \quad \textit{for all } s \in \mathcal{S},$$

where $r^\pi \in \mathbb{R}^S$ has entries $r^\pi_s = \sum_a \pi(a \mid s)R(s, a)$.

Proof. For a unichain policy, the ergodic theorem for finite Markov chains gives

$$\lim_{T\to\infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}_\pi[R(S_t, A_t) \mid S_0 = s] = \sum_{s'} \mu^\pi(s') r^\pi_{s'} = (\mu^\pi)^\top r^\pi,$$

regardless of the initial state s. This is because $(P^\pi)^t_{ss'} \to \mu^\pi(s')$ as $t \to \infty$ for all s (convergence to stationarity in the unichain case), so the Cesaro average converges to the stationary expectation. □

Definition 4.4 (Optimal Gain)**.** The *optimal gain* is $\rho^* = \sup_\pi \rho^\pi$. For unichain MDPs, the supremum is over all stationary policies (the gain is state-independent).

Example 4.5 (Average Reward of a Simple MDP)**.** Consider $\mathcal{S} = \{1, 2\}$, $\mathcal{A} = \{a\}$ (single action), with $P(1 \mid 1, a) = 0.3$, $P(2 \mid 1, a) = 0.7$, $P(1 \mid 2, a) = 0.6$, $P(2 \mid 2, a) = 0.4$. Rewards: $R(1, a) = 5$, $R(2, a) = 1$.

The stationary distribution satisfies $\mu(1) \cdot 0.7 = \mu(2) \cdot 0.6$, so $\mu(1)/\mu(2) = 6/7$. Normalizing: $\mu(1) = 6/13$, $\mu(2) = 7/13$.

The gain is $\rho = \mu(1)R(1, a) + \mu(2)R(2, a) = \frac{6}{13} \cdot 5 + \frac{7}{13} \cdot 1 = \frac{30+7}{13} = \frac{37}{13} \approx 2.85$.

5 The Gain-Bias Equations

The average reward analogue of the Bellman equations involves two quantities: the gain ρ^π (a scalar) and the *bias* h^π (a vector), also called the *relative value function* or *differential value function*.

Definition 5.1 (Bias / Relative Value Function)**.** For a unichain policy π with gain ρ^π, the *bias* (or *relative value function*) is

$$h^\pi(s) = \mathbb{E}_\pi\left[\sum_{t=0}^{\infty} (R(S_t, A_t) - \rho^\pi) \;\middle|\; S_0 = s\right].$$

Remark 5.2. The series in the bias definition converges because $R(S_t, A_t) - \rho^\pi$ has zero mean under the stationary distribution, and the deviations decay geometrically due to ergodicity. Formally, $|\mathbb{E}_\pi[R(S_t, A_t) - \rho^\pi \mid S_0 = s]| \le C\lambda^t$ for some $C > 0$ and $\lambda \in (0, 1)$ depending on the mixing rate of the chain.

Theorem 5.3 (Gain-Bias Equations for Policy Evaluation)**.** *For a unichain policy π, the gain $\rho^\pi \in \mathbb{R}$ and bias $h^\pi \in \mathbb{R}^S$ satisfy the* gain-bias equations *(or* Poisson equations*):*

$$\rho^\pi + h^\pi(s) = \sum_{a\in\mathcal{A}} \pi(a \mid s) \left[R(s,a) + \sum_{s'\in\mathcal{S}} P(s' \mid s,a)\, h^\pi(s') \right] \quad \textit{for all } s \in \mathcal{S}. \tag{1}$$

In matrix form: $\rho^\pi \mathbf{1} + h^\pi = r^\pi + P^\pi h^\pi$, or equivalently $(I - P^\pi)h^\pi = r^\pi - \rho^\pi \mathbf{1}$.

Proof. Starting from the bias definition, for any s:

$$\begin{aligned}
h^\pi(s) &= \mathbb{E}_\pi\left[\sum_{t=0}^{\infty} (R(S_t, A_t) - \rho^\pi) \;\middle|\; S_0 = s \right] \\
&= \mathbb{E}_\pi[R(S_0, A_0) - \rho^\pi \mid S_0 = s] + \mathbb{E}_\pi\left[\sum_{t=1}^{\infty} (R(S_t, A_t) - \rho^\pi) \;\middle|\; S_0 = s \right] \\
&= \sum_a \pi(a \mid s) R(s,a) - \rho^\pi + \sum_{s'} P^\pi_{ss'}\, \mathbb{E}_\pi\left[\sum_{t=0}^{\infty} (R(S_t, A_t) - \rho^\pi) \;\middle|\; S_0 = s' \right] \\
&= r^\pi_s - \rho^\pi + \sum_{s'} P^\pi_{ss'}\, h^\pi(s').
\end{aligned}$$

The third equality uses the Markov property and stationarity. Rearranging gives (1). □

Remark 5.4. The gain ρ^π is unique, but the bias h^π is determined only up to an additive constant: if (ρ, h) solves (1), then so does $(\rho, h + c\mathbf{1})$ for any $c \in \mathbb{R}$. This is because $(I - P^\pi)\mathbf{1} = 0$ (the rows of P^π sum to 1). To obtain a unique solution, one typically imposes a normalization such as $\sum_s \mu^\pi(s) h^\pi(s) = 0$ or $h^\pi(s_0) = 0$ for a reference state s_0.

Gain-Bias Equations (Poisson Equations)

For a unichain policy π with gain ρ^π and bias h^π:

$$\rho^\pi + h^\pi(s) = \sum_a \pi(a \mid s) \left[R(s,a) + \sum_{s'} P(s' \mid s,a)\, h^\pi(s') \right], \quad s \in \mathcal{S}.$$

Compared to the discounted Bellman equation $V^\pi(s) = \sum_a \pi(a \mid s)[R(s,a) + \gamma \sum_{s'} P(s' \mid s,a) V^\pi(s')]$, the gain ρ^π replaces the discount factor's role: it is the constant rate subtracted from immediate rewards to make the remaining value bounded.

Example 5.5 (Computing Gain and Bias)**.** Return to Example 4.5: $\mathcal{S} = \{1, 2\}$, $r = (5, 1)^\top$, $P = \left(\begin{smallmatrix} 0.3 & 0.7 \\ 0.6 & 0.4 \end{smallmatrix}\right)$, and we computed $\rho = 37/13$.

The gain-bias equations are:

$$\begin{aligned}
\frac{37}{13} + h(1) &= 5 + 0.3h(1) + 0.7h(2), \\
\frac{37}{13} + h(2) &= 1 + 0.6h(1) + 0.4h(2).
\end{aligned}$$

Rearranging: $0.7h(1) - 0.7h(2) = 5 - 37/13 = 28/13$ and $-0.6h(1) + 0.6h(2) = 1 - 37/13 = -24/13$.

From the first: $h(1) - h(2) = 28/(13 \cdot 0.7) = 40/13$. Setting $h(2) = 0$ (normalization): $h(1) = 40/13 \approx 3.08$.

The bias $h(1) > h(2)$ reflects that state 1 is "better than average" (reward $5 >$ gain $37/13 \approx 2.85$), so starting there yields a transient bonus.

6 Average Reward Optimality Equations

Theorem 6.1 (Average Reward Bellman Optimality Equations)**.** *For a unichain MDP, there exist a scalar ρ^* (the optimal gain) and a vector $h^* \in \mathbb{R}^S$ (the optimal bias) satisfying:*

$$\rho^* + h^*(s) = \max_{a \in \mathcal{A}} \left[R(s,a) + \sum_{s' \in \mathcal{S}} P(s' \mid s,a)\, h^*(s') \right] \quad \text{for all } s \in \mathcal{S}. \tag{2}$$

The optimal gain ρ^* is unique. The optimal bias h^* is unique up to an additive constant.

Proof. The proof uses the connection to discounted MDPs via the Laurent series expansion (developed in Theorem 7.1). We provide an outline.

Step 1: Existence. Consider the discounted optimal value function V_γ^* as a function of γ. For a unichain MDP, define

$$\rho^* = \lim_{\gamma \uparrow 1} (1-\gamma) V_\gamma^*(s)$$

(this limit exists and is independent of s for unichain MDPs) and

$$h^*(s) = \lim_{\gamma \uparrow 1} [V_\gamma^*(s) - V_\gamma^*(s_0)]$$

for some reference state s_0. Taking the limit $\gamma \uparrow 1$ in the discounted Bellman optimality equation $V_\gamma^*(s) = \max_a [R(s,a) + \gamma \sum_{s'} P(s' \mid s,a) V_\gamma^*(s')]$ yields (2).

Step 2: Uniqueness of ρ^*. Suppose (ρ_1, h_1) and (ρ_2, h_2) both satisfy (2) with $\rho_1 > \rho_2$. Let d_2 be the greedy policy w.r.t. h_2, i.e., the maximizer in (2) for (ρ_2, h_2). Then for all s: $\rho_1 + h_1(s) \geq R(s, d_2(s)) + \sum_{s'} P(s' \mid s, d_2(s)) h_1(s') = \rho_2 + h_2(s) + \sum_{s'} P(s' \mid s, d_2(s))[h_1(s') - h_2(s')]$. Summing with the stationary distribution μ^{d_2} and using $\sum_s \mu^{d_2}(s) \sum_{s'} P(s' \mid s, d_2(s)) f(s') = \sum_{s'} \mu^{d_2}(s') f(s')$ for any f, we get a contradiction unless $\rho_1 = \rho_2$. □

Definition 6.2 (Average Reward Greedy Policy)**.** Given a solution (ρ^*, h^*) to (2), the *average reward optimal policy* is

$$\pi^*(s) = \arg\max_{a \in \mathcal{A}} \left[R(s,a) + \sum_{s'} P(s' \mid s,a)\, h^*(s') \right].$$

Theorem 6.3 (Optimality of the Greedy Policy)**.** *For a unichain MDP, the greedy policy π^* with respect to h^* achieves the optimal gain: $\rho^{\pi^*} = \rho^*$.*

Proof. Since π^* achieves the max in (2), h^* satisfies the gain-bias equation $\rho^* + h^*(s) = R(s, \pi^*(s)) + \sum_{s'} P(s' \mid s, \pi^*(s)) h^*(s')$ for all s. Multiplying by $\mu^{\pi^*}(s)$ and summing: $\rho^* = \sum_s \mu^{\pi^*}(s) R(s, \pi^*(s)) = \rho^{\pi^*}$. □

Gain-Bias Equations and R-Learning

The R-learning algorithm directly estimates ρ^* and h^* using stochastic approximation. The update rules are:

$$h(s_t, a_t) \leftarrow h(s_t, a_t) + \alpha_t [R(s_t, a_t) - \hat{\rho} + \max_{a'} h(s_{t+1}, a') - h(s_t, a_t)],$$
$$\hat{\rho} \leftarrow \hat{\rho} + \beta_t [R(s_t, a_t) + \max_{a'} h(s_{t+1}, a') - \max_{a'} h(s_t, a') - \hat{\rho}].$$

This is the average-reward analogue of Q-learning, with the gain estimate $\hat{\rho}$ replacing the discount factor.

7 Connection to the Discounted Criterion

The average reward criterion can be viewed as the limiting case of the discounted criterion as $\gamma \to 1$.

Theorem 7.1 (Discount-to-Average Limit)**.** *For a unichain MDP with optimal discounted value V_γ^* and optimal average reward ρ^*:*

1. $\lim_{\gamma\uparrow 1}(1-\gamma)V_\gamma^*(s) = \rho^*$ *for all* $s \in \mathcal{S}$.
2. *The limit* $h^*(s) = \lim_{\gamma\uparrow 1}[V_\gamma^*(s) - V_\gamma^*(s_0)]$ *exists for any reference state* s_0 *and satisfies the average reward Bellman optimality equation* (2).

Proof. **Part 1:** The discounted value satisfies $\|V_\gamma^*\|_\infty \le R_{\max}/(1-\gamma)$, so $(1-\gamma)\|V_\gamma^*\|_\infty \le R_{\max}$, and $(1-\gamma)V_\gamma^*$ is bounded.

For a fixed deterministic policy d, the discounted value is $V_\gamma^d = (I - \gamma P^d)^{-1} r^d$. Using the Neumann series: $(I-\gamma P^d)^{-1} = \sum_{t=0}^\infty \gamma^t (P^d)^t$.

For a unichain policy d, $(P^d)^t \to \mathbf{1}(\mu^d)^\top$ as $t \to \infty$, where μ^d is the stationary distribution. Therefore:

$$\begin{aligned}(1-\gamma)V_\gamma^d(s) &= (1-\gamma)\sum_{t=0}^\infty \gamma^t [(P^d)^t r^d]_s \\ &\to (\mu^d)^\top r^d = \rho^d \quad \text{as } \gamma \to 1.\end{aligned}$$

The interchange of limit and summation is justified by dominated convergence (the partial sums are bounded by $R_{\max}$).

Since $V_\gamma^*(s) = \max_d V_\gamma^d(s)$ and the max is over finitely many deterministic policies:

$$\lim_{\gamma\to 1}(1-\gamma)V_\gamma^*(s) = \max_d \lim_{\gamma\to 1}(1-\gamma)V_\gamma^d(s) = \max_d \rho^d = \rho^*.$$

The interchange of max and limit is valid because the max is over a finite set.

Part 2: The centered function $\tilde{V}_\gamma(s) = V_\gamma^*(s) - V_\gamma^*(s_0)$ satisfies $\|\tilde{V}_\gamma\|_\infty \le 2R_{\max}/(1-\gamma)$, but the span is bounded: $\mathrm{sp}(\tilde{V}_\gamma) = \mathrm{sp}(V_\gamma^*)$.

For a unichain MDP, $\mathrm{sp}(V_\gamma^*)$ remains bounded as $\gamma \to 1$ (this uses the span contraction from Lesson 43). By compactness (Bolzano-Weierstrass on $\mathbb{R}^S$), there exists a convergent subsequence, and one can show the limit satisfies (2) by passing to the limit in the discounted Bellman equation. Uniqueness of ρ^* ensures the full limit exists. □

Example 7.2 (Discounted to Average Reward)**.** For the MDP in Example 4.5, the discounted value under the single policy with $\gamma = 0.99$ is $V_\gamma(1) = (I - 0.99P)^{-1} r$. With $r = (5,1)^\top$ and $P = \left(\begin{smallmatrix} 0.3 & 0.7 \\ 0.6 & 0.4 \end{smallmatrix}\right)$:

$$I - 0.99P = \begin{pmatrix} 0.703 & -0.693 \\ -0.594 & 0.604 \end{pmatrix}.$$

The determinant is $0.703 \cdot 0.604 - 0.693 \cdot 0.594 = 0.4246 - 0.4117 = 0.0129$. Solving: $V_{0.99}(1) = (0.604 \cdot 5 + 0.693 \cdot 1)/0.0129 = 3.713/0.0129 \approx 287.8$.

Then $(1-0.99)V_{0.99}(1) = 0.01 \cdot 287.8 \approx 2.88 \approx 37/13 = \rho$. As $\gamma \to 1$, the convergence becomes exact.

Average Reward Requires Ergodicity

The limit $(1-\gamma)V_\gamma^*(s) \to \rho^*$ independent of s relies on the unichain assumption. For multichain MDPs, the limit can depend on the initial state, and the average reward optimality equations are more complex. In such cases, additional structure (e.g., communicating MDP assumption) or state-dependent gain is needed.

8 Relative Value Iteration

Value iteration cannot be directly applied to the average reward setting because there is no contraction: the Bellman operator $(\mathcal{T}_{avg}h)(s) = \max_a[R(s,a) + \sum_{s'} P(s' \mid s,a)h(s')]$ (with no discount) satisfies $\|\mathcal{T}_{avg}u - \mathcal{T}_{avg}v\|_\infty \leq \|u - v\|_\infty$ (non-expansive, but not a contraction). However, *relative value iteration* resolves this by centering the iterates.

Definition 8.1 (Relative Value Iteration)**.** *Relative value iteration* (RVI) generates sequences $\{h^{(n)}\}$ and $\{\rho^{(n)}\}$ by:

1. **Initialize:** Choose $h^{(0)} \in \mathbb{R}^S$.
2. **Iterate:** For $n = 0, 1, 2, \ldots$,

$$\begin{aligned} w^{(n+1)}(s) &= \max_{a\in\mathcal{A}} \left[R(s,a) + \sum_{s'} P(s' \mid s,a)\, h^{(n)}(s') \right], \\ \rho^{(n+1)} &= w^{(n+1)}(s_0) \quad \text{(for a fixed reference state } s_0\text{)}, \\ h^{(n+1)}(s) &= w^{(n+1)}(s) - \rho^{(n+1)}. \end{aligned}$$

Remark 8.2. The subtraction $h^{(n+1)}(s) = w^{(n+1)}(s) - w^{(n+1)}(s_0)$ ensures $h^{(n+1)}(s_0) = 0$ at every iteration, preventing the iterates from diverging. This is necessary because the undiscounted Bellman operator shifts values by the gain at each step.

Theorem 8.3 (Convergence of Relative Value Iteration)**.** *For a unichain MDP, relative value iteration converges: $\rho^{(n)} \to \rho^*$ and $h^{(n)} \to h^*$ (up to the normalization $h^*(s_0) = 0$).*

Proof. The span contraction from Lesson 43 is the key tool. Since $\mathrm{sp}(\mathcal{T}_{avg}u - \mathcal{T}_{avg}v) \leq \mathrm{sp}(u - v)$ for the undiscounted operator (with contraction coefficient 1), the convergence analysis requires the stronger result that the span contraction coefficient is strictly less than 1 for unichain MDPs, a result known as the *span condition*. We outline the argument.

Define $\sigma = \max_{a_1,a_2} \min_{s'}[P(s' \mid \cdot, a_1) + P(s' \mid \cdot, a_2)]$ (the *scrambling coefficient*). For a communicating MDP, $\sigma > 0$, and the span contraction coefficient of $\mathcal{T}_{avg}$ is at most $1 - \sigma < 1$. Standard results then give convergence in the span seminorm, which implies convergence of $\rho^{(n)}$ and convergence of $h^{(n)}$ up to a constant.

A complete proof can be found in Puterman [2014, Chapter 8]. □

Example 8.4 (Relative Value Iteration)**.** For the MDP of Example 4.5 with a single action, starting from $h^{(0)} = (0,0)^\top$ and $s_0 = 2$:

Iteration 1: $w^{(1)}(1) = 5 + 0.3 \cdot 0 + 0.7 \cdot 0 = 5$, $w^{(1)}(2) = 1 + 0.6 \cdot 0 + 0.4 \cdot 0 = 1$. $\rho^{(1)} = w^{(1)}(2) = 1$. $h^{(1)} = (5-1, 1-1)^\top = (4,0)^\top$.

Iteration 2: $w^{(2)}(1) = 5 + 0.3 \cdot 4 + 0.7 \cdot 0 = 6.2$, $w^{(2)}(2) = 1 + 0.6 \cdot 4 + 0.4 \cdot 0 = 3.4$. $\rho^{(2)} = 3.4$. $h^{(2)} = (6.2 - 3.4, 0)^\top = (2.8, 0)^\top$.

Iteration 3: $w^{(3)}(1) = 5 + 0.3 \cdot 2.8 = 5.84$, $w^{(3)}(2) = 1 + 0.6 \cdot 2.8 = 2.68$. $\rho^{(3)} = 2.68$. $h^{(3)} = (3.16, 0)^\top$.

The gain sequence $1, 3.4, 2.68, \ldots$ oscillates toward $\rho^* = 37/13 \approx 2.846$, and the bias $h^{(n)}(1) \to 40/13 \approx 3.077$.

9 Blackwell Optimality

Blackwell optimality provides a remarkable connection between the discounted and average reward criteria.

Definition 9.1 (Blackwell Optimality)**.** A deterministic stationary policy d^* is *Blackwell optimal* if there exists $\gamma_0 \in (0,1)$ such that d^* is γ-discount optimal for all $\gamma \in (\gamma_0, 1)$. That is, $V_\gamma^{d^*}(s) = V_\gamma^*(s)$ for all s and all $\gamma > \gamma_0$.

Theorem 9.2 (Blackwell's Theorem)**.** *For any finite MDP, a Blackwell optimal policy exists. Moreover, every Blackwell optimal policy is average-reward optimal.*

Proof. **Existence:** For each deterministic stationary policy d, the discounted value $V_\gamma^d(s)$ is a rational function of γ (since $V_\gamma^d = (I - \gamma P^d)^{-1} r^d$ and each entry of the inverse is a ratio of polynomials in γ via Cramer's rule).

For two policies d_1, d_2, the difference $V_\gamma^{d_1}(s) - V_\gamma^{d_2}(s)$ is a rational function of γ and therefore has finitely many zeros in $(0, 1)$ (unless identically zero). This means the set of γ values where d_1 and d_2 swap in terms of which is better is finite.

Since there are finitely many policies (A^S total), the union of all such swap points is finite. Therefore, there exists $\gamma_0 < 1$ such that for all $\gamma > \gamma_0$, the set of discount-optimal policies does not change. Any policy that is optimal for all $\gamma > \gamma_0$ is Blackwell optimal.

Average-reward optimality: If d^* is Blackwell optimal, then $(1-\gamma)V_\gamma^{d^*}(s) = (1-\gamma)V_\gamma^*(s)$ for $\gamma > \gamma_0$. Taking $\gamma \to 1$: $\rho^{d^*} = \lim(1-\gamma)V_\gamma^{d^*}(s) = \lim(1-\gamma)V_\gamma^*(s) = \rho^*$. □

Blackwell Optimality

A Blackwell optimal policy exists for every finite MDP. It is simultaneously:

- γ-discount optimal for all γ sufficiently close to 1,
- average-reward optimal ($\rho^{d^*} = \rho^*$).

Blackwell optimality is the strongest notion of optimality for infinite-horizon MDPs, subsuming both discounted and average reward optimality.

Example 9.3 (Blackwell Optimality in a Two-Action MDP)**.** Consider $\mathcal{S} = \{1\}$ (single state), $\mathcal{A} = \{a, b\}$ with $R(1, a) = 2$, $R(1, b) = 1$. Since there is only one state, both actions lead back to state 1. For any γ, the optimal policy is $\pi^*(1) = a$ (the higher-reward action). This policy is trivially Blackwell optimal with $\gamma_0 = 0$: it is optimal for all discount factors.

For a more interesting example, consider $\mathcal{S} = \{1, 2\}$ with:

s	a	$R(s,a)$	$P(1 \mid s,a)$	$P(2 \mid s,a)$
1	a	0	0	1
1	b	1	1	0
2	a	3	1	0
2	b	3	1	0

Policy d_1: (b, a) gives $P^{d_1} = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 0 \end{smallmatrix}\right)$, $r^{d_1} = (1, 3)^\top$. Stationary distribution: $\mu = (1, 0)$, gain $\rho^{d_1} = 1$.

Policy d_2: (a, a) gives $P^{d_2} = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, $r^{d_2} = (0, 3)^\top$. Stationary distribution: $\mu = (1/2, 1/2)$, gain $\rho^{d_2} = 3/2$.

Policy d_2 has higher average reward. For the discounted case, $V_\gamma^{d_2}(1) = \frac{3\gamma}{1-\gamma^2} = \frac{3\gamma}{(1-\gamma)(1+\gamma)}$ and $V_\gamma^{d_1}(1) = \frac{1}{1-\gamma}$. We have $V_\gamma^{d_2}(1) > V_\gamma^{d_1}(1)$ iff $\frac{3\gamma}{1+\gamma} > 1$ iff $\gamma > 1/2$. So d_2 is Blackwell optimal with $\gamma_0 = 1/2$.

10 Exercises

Exercise 10.1 (Gain and Bias Computation)**.** $(*)$
Consider a two-state MDP with $\mathcal{S} = \{1, 2\}$, $\mathcal{A} = \{a, b\}$:

s	a	$R(s,a)$	$P(1 \mid s,a)$	$P(2 \mid s,a)$
1	a	4	0.5	0.5
1	b	2	0.8	0.2
2	a	1	0.3	0.7
2	b	3	0.6	0.4

(a) For the policy $d(1) = a$, $d(2) = b$, compute the stationary distribution μ^d and the gain ρ^d.

(b) Solve the gain-bias equations for h^d (with normalization $h^d(2) = 0$).

Exercise 10.2 (Unichain Classification). $(*)$
For the MDP in Exercise 1, check whether each of the four deterministic policies induces a unichain Markov chain. Is the MDP unichain?

Exercise 10.3 (Average Reward Optimality Equations). $(**)$
For the MDP in Exercise 1:

(a) Write down the average reward Bellman optimality equations (2).

(b) Solve for ρ^* and h^* (with $h^*(2) = 0$) by trying each candidate optimal policy.

(c) Verify by computing the gain of all four deterministic policies.

Exercise 10.4 (Discount-to-Average Convergence). $(**)$
For the MDP in Example 4.5:

(a) Compute V_γ^* for $\gamma = 0.9, 0.99, 0.999$.

(b) Verify that $(1-\gamma)V_\gamma^*(s) \to \rho^* = 37/13$ for each s.

(c) Compute $V_\gamma^*(1) - V_\gamma^*(2)$ for each γ and verify convergence to $h^*(1) - h^*(2) = 40/13$.

Exercise 10.5 (Blackwell Optimality Threshold). $(***)$
For the MDP in Example 9.3:

(a) Compute $V_\gamma^d(s)$ for all four deterministic policies and both states, as functions of γ.

(b) Determine the Blackwell threshold γ_0 (the largest γ at which the optimal policy changes).

(c) Verify that the Blackwell optimal policy is indeed average-reward optimal.

Exercise 10.6 (Multichain Counterexample). $(***)$
Construct a three-state, two-action MDP that is multichain (at least one deterministic policy has two recurrent classes) and show that the average reward can depend on the initial state. Specifically, find a policy π and two states s_1, s_2 such that $\rho^\pi(s_1) \neq \rho^\pi(s_2)$.

Hint: Use absorbing states to create separate recurrent classes with different reward rates.

11 Summary

- The *average reward criterion* evaluates policies by their long-run reward rate $\rho^\pi = \lim_{T\to\infty} \frac{1}{T}\mathbb{E}[\sum_{t=0}^{T-1} R(S_t$ eliminating the discount factor as a hyperparameter.
- For *unichain* MDPs (every policy induces a chain with a single recurrent class), the gain ρ^π is independent of the initial state and equals the expected reward under the stationary distribution: $\rho^\pi = (\mu^\pi)^\top r^\pi$.

- The *gain-bias equations* (Poisson equations) $\rho^\pi + h^\pi(s) = \sum_a \pi(a \mid s)[R(s,a) + \sum_{s'} P(s' \mid s,a)h^\pi(s')]$ are the average-reward analogue of the discounted Bellman equations. The bias $h^\pi(s)$ measures the transient advantage of starting in state s.
- The bias is unique only up to an additive constant, reflecting that only *relative* values matter in the average reward setting.
- The average reward Bellman optimality equation $\rho^* + h^*(s) = \max_a[R(s,a) + \sum_{s'} P(s' \mid s,a)h^*(s')]$ characterizes the optimal gain and bias.
- *Relative value iteration* centers iterates at a reference state to prevent divergence, enabling iterative solution of the average reward optimality equations.
- The discounted and average reward criteria are connected: $(1-\gamma)V^*_\gamma \to \rho^*$ as $\gamma \to 1$, and the centered discounted values converge to the bias.
- A *Blackwell optimal* policy is simultaneously discount-optimal for all γ sufficiently close to 1 and average-reward optimal. Blackwell optimal policies exist for all finite MDPs.

12 Further Reading

- **Lesson on TD Learning** extends the ideas from this lesson to the model-free setting, where the transition kernel is unknown and the agent learns from sampled trajectories.
- **Lesson on Stochastic Approximation** (Lesson 19) provides the convergence framework for R-learning and other average-reward RL algorithms.
- Puterman [2014, Chapters 8–10] provides the definitive treatment of average reward MDPs, including multichain theory, sensitive-discount optimality, and Laurent series expansions.
- Blackwell [1962] is the original paper establishing Blackwell optimality; it remains concise and readable.
- Howard [1960] introduced the gain-bias framework and developed policy iteration for average reward, establishing the computational foundations.
- Bertsekas [2012, Chapter 7] treats average cost problems with connections to stochastic shortest path formulations and semi-Markov decision processes.
- Derman [1970] provides an accessible treatment of average reward theory with applications to queuing and inventory problems.

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